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Body of European Regulators for Electronic Communications

BEREC External Study on Data Centers

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Motivation and organisation

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Introduction

- BEREC considered important to understand how data centers deployment is evolving, main actors, dynamics, and implications on connectivity.
- For this reason, we commissioned an external study that was directed by a BEREC Steering Group composed of experts from several NRAs with the support of the BEREC Office, focused on:
  - Preparing tender specifications
  - Selection of the most adequate offer
  - Meeting regularly with contractor
  - Review and comments on drafts prepared by the contractor
- Tender launched Feb. 2025, 7 proposals received
- Contractors: PwC and Poznań Supercomputing and Networking Centre (PSNC)
- Work prepared from June 2025 to December 2025
- Published in February 2026
- The report does not necessarily reflect BEREC views.

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Objectives, methology and deliverables

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Project Objectives & Methodology

Research objectives:

1. The current state of the DCs market in Europe
2. Mapping of DC Interdependencies with ECN Infrastructure
3. Analysis of Competition Dynamics in a Converging Ecosystem
4. Forward-Looking Insights for Regulatory Adaptation
5. Comprehensive Assessment of DC Sustainability

Mixed-method research approach included a comprehensive desk research, survey and interviews:



Desk research

Concentrated on regulatory policies, market trends, and sustainability standards.



Survey

Conducted with over 40 European DS operators from telecom, cloud, and other sectors, focusing on connectivity, energy use, and cooling systems.



Interviews

Comprehensive 18 interviews with representatives from data center operators, telecommunications firms, regulatory agencies, and academic institutions.

Key project outcomes

1. **Final study report**

2. **Data Centre Database:**

We compiled a **DC database** including the following information:

- **Name, location,** and year of entry into operation
- Date of last expansion/modification
- Main purpose of the data centre
  (e.g., colocation, cloud provider, telecommunications, etc.)
- **Energy consumption** and **connectivity details**
  (fixed, mobile, submarine cables, redundancy levels)
- **Relationship with ECN providers**
  (e.g., direct ownership by telecom operators)
- Technical parameters
  (building size, number of cabinets, cooling technology, security)
- Certification level
- **Ownership structure** and level of openness to 3rd-party services

Key Highlights:

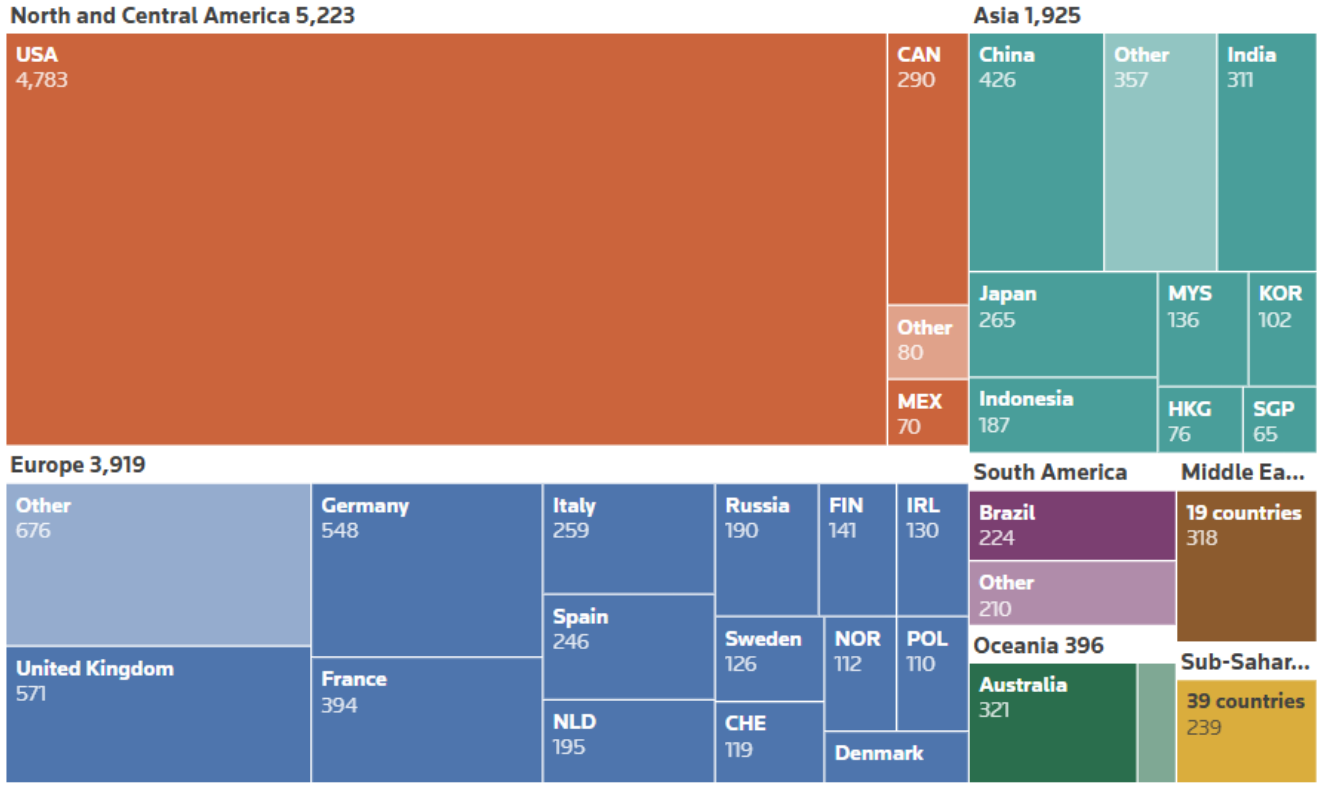
- Presence in 30 Countries
- 309 Cities
- 109 Unique Owners

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DC Database

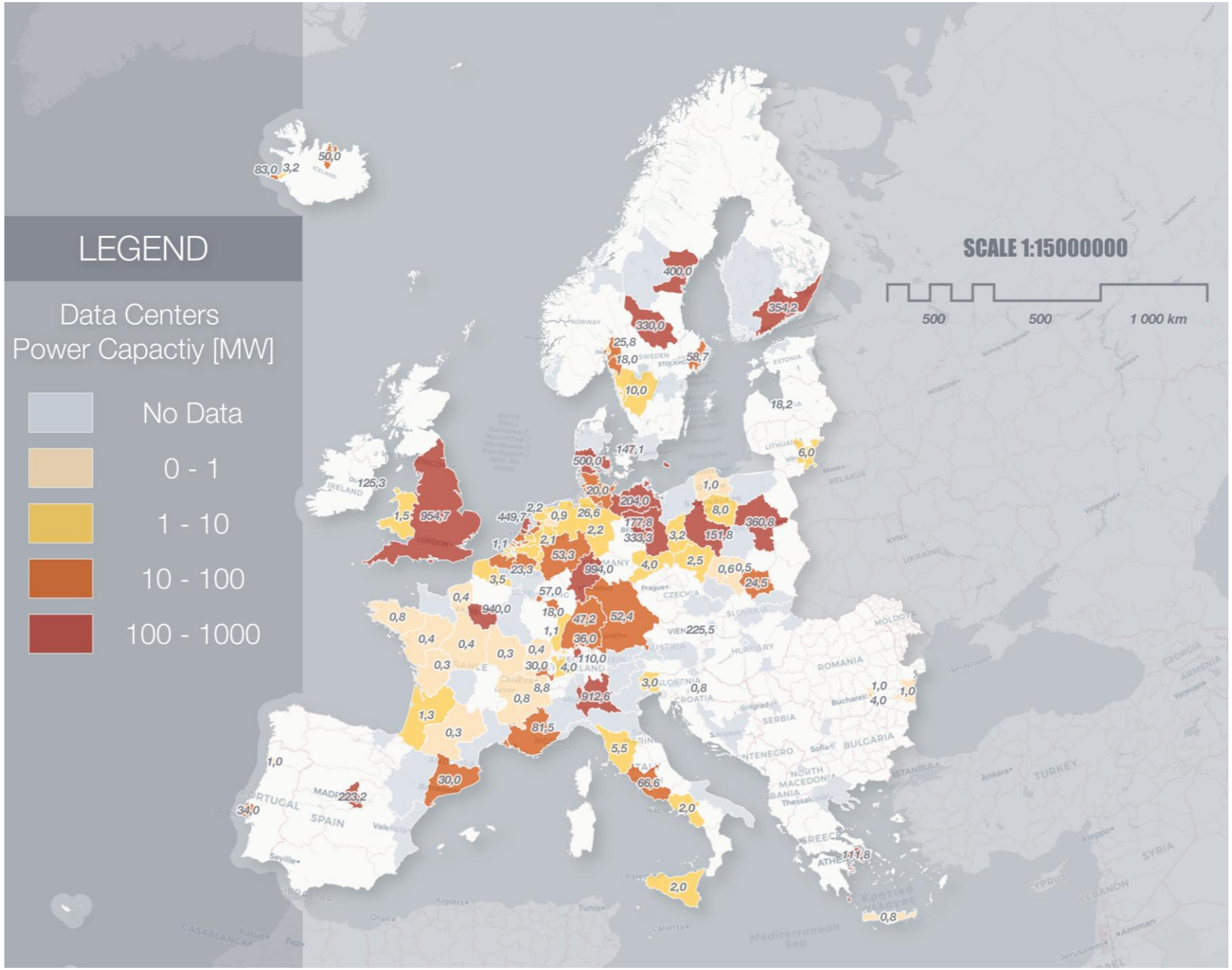
A view on data centers in the world



Note: Includes operational, under-construction and planned data centers, as well as sites where land has been acquired or status is uncertain; excludes canceled and decommissioned ones. Each building on a multi-building campus counts separately; multiple operators in the same building count once. Countries with fewer than 10 data centers are grouped as "Other". Turkey is geographically and politically in both Europe and Asia, but is listed under Europe for this chart only. c.Data as of Sept. 29, 2026.

Source: Data Center Map

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DC Database – max power capacity of identified DCs per region

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Main study results

Geographic Distribution, Ownership Models, and Key Dependencies

Europe hosts

~3,000 data centres

(global >11,800)

Market size **valued**

USD 47.23B **(2024)** →

projected **USD** 97.30B

by 2030 (CAGR 12.8%).

Drivers

Power grid: availability and infrastructure; power pricing; connection lead-times

Connectivity: access to high-capacity backbones; diverse carrier connectivity; proximity to cloud ecosystems, Internet Exchange Points (IXPs) and customers to reduce latency

Cooling and water resources: water availability; operational cost and disclosure; regulatory and investor implications

Supply chains: hardware availability trade policy (tariffs! GPUs!); logistics; standards compliance

Environmental & Regulatory Dependencies: EU policies; local constraints and moratoria; fragmented national regimes

Other factors:
land availability;
skilled workforce shortages

Ownership models**: independent colocation**; **hyperscaler-owned**; **public/PPP**; **telco-owned**, and **hybrid/emerging (edge, sovereign/AI, quantum-ready)** — with different capex profiles, revenue mixes, and regulatory exposure.

Geographic Distribution

Geographic inequalities between N/W and S/E:

1

Capacity concentration

- **W/NE face saturation - FLAP-D (+Nordics)** hold **~60–65%** of capacity; S/CEE still under-represented: secondary hubs (Madrid, Berlin, Warsaw) scaling from small base
- **Network infrastructure**: W/NE host most cable landings and top **IXPs**; S/CEE peering is improving but thinner; *cable landings in Spain, Portugal stimulate DC market development*
- **Power-grids:** Legacy hubs face **7–13 years** for HV connections *(or are stoped by moratoria)*; S/CEE sites **3–5 years** (but weaker redundancy)

2

Skills ecosystem maturity

- W/NE benefits from mature ecosystems;
- S/CEE developing necessary technical capacity.

3

Submarine cables: strategic digital arteries

- W/NE — transparent, more predictable permitting & incentives;
- S/CEE — decentralized procedures → higher uncertainty and delays.

4

Key actions to move forward

Europe's DC expansion will follow: **power, interconnection, predictable permitting**;

Closing gaps needs:

- **policy:** harmonise policy across Europe
- **grid:** balancing the development, maturing S/CEE structures
- **ecosystem** co-investment: governments and European companies (energy, telco, DC) to act together
- **network:** low-latency network availability should follow the grid and ecosystem capabilities to enable DCs growth in promising locations

DC Interdependencies with ECNs and DCs

1

ECNs' strategic investments in DCs

- **Key drivers**: low-latency service demand, revenue diversification; network efficiency, digital sovereignty (Data Governance Act/DataAct, GAIA-X), and strategic control amid the AI build-out.
- **Focus**: edge/MEC nodes; collocation facilities & cloud on-ramps; terrestrial & subsea backbones (e.g. Project Visegrád); renewable and energy-efficient infrastructure; sovereign-cloud/PPPs, AI-ready high-density sites.

2

Implications for long-term infrastructure development

- **Economic/competition**: scarcity in primary hubs, shift to secondary markets; case for co-investment/platform plays.
- **Infrastructure/sovereignty**: network-compute convergence; faster edge/MEC adoption; standards (EUCS/GAIA-X) shaping "trusted" infrastructure.

3

Key gaps to close

- **Financial/market gaps**: high capex, grid/HV-connection delays, telco capex diverted to 5G/fibre → need JVs/long-term leases and predictable power access.
- **Operational gaps**: skills shortages; ECN–DC ops integration challenges; meeting high-availability/efficiency targets.
- **Regulatory gaps**: fragmented permitting and overlapping EU frameworks) raising compliance burden.

Key aspects on connectivity for DCs

Hyperscalers bypass legacy networks	Tech giants are building private transoceanic cables (e.g., Google's "Equiano," "Dunant," Meta's "2Africa") to directly link data centres, reducing dependency on telecom-operated infrastructure.
AI-driven surge in interconnection demand	The AI boom is driving record investments in data centres (global spending expected to exceed $250 B by 2028) and creating pressure for high-capacity, low-latency interconnects.
Fiber and orchestration innovation	Advances in multi-core and hollow-core fiber offer up to 3× bandwidth increases, while software-defined networking and optical orchestration improve traffic management, flexibility, and resilience of inter-DC links.
Security and privacy as design priorities	Interconnection strategies increasingly integrate end-to-end encryption, quantum-safe protocols, and zero-trust architectures to mitigate data interception risks and ensure compliance with regional regulations (GDPR, NIS2, DORA).
Lack of global contingency planning	There is no coordinated international framework to preserve global connectivity during large-scale disruptions, from hybrid warfare, seismic activity, or sabotage. Response capacity remains highly fragmented, over 95% of subsea cables remain privately owned by consortia or hyperscalers.

Environmental Sustainability and Energy Consumption

The EU Energy Efficiency Directive (EED), part of the "Fit for 55" package, is the main driver for sustainability, aiming for climate neutrality by 2050

1 Mandatory Reporting

DCs with a total rated IT power of >500kW must report their energy performance annually to a European database.

2 Key Performance Indicators (KPIs)

- PUE (Power Usage Effectiveness) – optimization, highly depends on server and network technology.
- WUE (Water Usage Effectiveness) – depends on cooling technology and geographical location.
- ERF (Energy Reuse Factor) – depends on the availability of an appropriate district heating network.
- REF (Renewable Energy Factor) - Power Purchase Agreements (PPAs), owning one's own PV systems and wind turbines is difficult to implement.

3 PUE Targets

- New DCs (from July 2026) must achieve **PUE** ≤1.2 – it could be challenging because of servers, storage, network systems technology and geographical location
- Existing DCs must reach PUE ≤1.3 by July 2030 – challenging for old Data Centres due to cooling circles design – realistic value is ~1.5
- PUE =~ 1.05 – only achievable to Data Centers with dedicated technology (full DLC and more)

4 Waste Heat Utilization (ERF)

DCs >1MW must utilize waste heat for recovery applications, unless infeasible

- Germany sets a target of ≥20% **ERF by July 2028 –** it is challenging to achieve because too many heat, insufficient demand, and unsuitable temperatures (<40 oC).

5 Renewable Sourcing

Germany mandates that DCs source **100% of electricity from renewables by January 1, 2027;**
Operators primarily use **Power Purchase Agreements (PPAs)** to achieve this.

Environmental Sustainability and Energy Consumption

1

Water Usage

- The industry average WUE reported in 2023 was 0.31 l/kWh = ~75% of the CNDCP target of **0.4 l/kWh** for water-stressed areas
- **Direct Liquid Cooling (DLC)** systems use closed loops, minimizing water demand

2

E-Waste and Circular Economy

- **"6R"** principles (Rethink, Refuse, Reduce, Reuse,Recycle, Recover/Repair) guide circular strategies
- CNDCP mandates that operators must assess **100% of used server equipment** for reuse, repair, or recycling

3

Heat Reuse

- Integration with district heating networks is essential for heat recovery – in summer almost no heat is required, in winter there is free heat from the combined heat and power plant
- Example: DCs in Berlin are integrating 8MW of waste heat capacity into local district heating networks - infrastructure can serve dual purposes. Similar projects in Poznań, Stockholm focus on reusing waste heat for buildings and campuses.

Competition and Market Dynamics in Europe's Data Centre Market

1

Evolving Competitive Landscape

- Strong competition among telecom operators, US hyperscalers, and independent colocation providers.
- Hyperscalers lead new demand and investment, while telcos focus more on partnerships than ownership.
- Colocation providers stay competitive by entering secondary markets and adding services (as basic collocation commoditizes)

2

Consolidation and New Entrants

- The sector sees mergers, acquisitions, and private equity buyouts alongside new entrants like infrastructure funds and regional specialists.
- Large operators are growing, but the market stays fragmented continentally, with strong regional competition and ongoing investments in emerging markets such as Spain, Italy, and Poland.

Future Regulatory Trends in Europe's Data Centre and Cloud Market

1

Stricter and Proactive Regulation Ahead

- The EU is shifting to proactive rules focused on competition, energy efficiency, carbon neutrality, and digital sovereignty.
- Regulation is becoming more cross-sectoral, integrating energy, competition, digital, and industrial policy.

2

Competition, Interoperability, and Sovereignty

- New rules (e.g., the Data Act, Digital Markets Act) target cloud interoperability, reduce switching costs, and are aimed to curb hyperscalers dominance.
- The EU supports better interoperability and European cloud options.
- Data sovereignty is emphasized through.

3

Sustainability and Circular Economy as Core Requirements

- Data centres must meet EU Green Deal and Fit for 55 goals for climate neutrality by 2030.
- Transparency, compliance, and innovations such as advanced cooling and waste heat recovery will be key for competitiveness.
- Data centres increasingly seen as assets for grid flexibility, not just energy consumers.

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References

References

- **The report:**

https://www.berec.europa.eu/en/all-documents/berec/reports/external-study-on-data-centres

- **Tender technical specifications:**

https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/tender-details/docs/a7ab1654-9650-4ce1-8e09-94dda75bdae3-CN/TENDER%20SPECIFICATIONS_Part%202_Technical%20specifications_BEREC2025OP0002_BEREC%20External%20Study%20on%20data%20centres_V1.pdf

- **All tender details:**

https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/opportunities/tender-details/a7ab1654-9650-4ce1-8e09-94dda75bdae3-CN

- **BEREC Report on the entry of large content and application providers into the markets for electronic communications networks and services (BoR (24) 139):**

https://www.berec.europa.eu/en/all-documents/berec/reports/berec-report-on-the-entry-of-large-content-and-application-providers-into-the-markets-for-electronic-communications-networks-and-services

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Muchas gracias!

Muito obrigado!

Thank you!

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